

Maharashtra HSC Exam, 2019

Chemistry

Time-3 Hours

M.M- 70

General Instructions:

1. All questions are compulsory.
2. Answers to the questions of Section-A, B, C and D should be written in the same answer book.
3. In Section –A there are 8 questions each of 1 marks.
4. In Section – B there are 7 questions each of 2 marks.
5. In Section- C there are 11 questions each of 3 marks.
6. In Section- D there are 3 questions each of 5 marks.
7. Draw neat, labelled diagrams and write balanced chemical equations wherever necessary.
8. Figures to the right indicate full marks.
9. Use of logarithmic table is allowed.
10. Every new question must be started on a new page.

Section-I

Question 1: A compound used as pistachio flavor in ice cream is _____. [1]

- (a) Vanilin (b) Acetophenone (c) Muscone (d) butyraldehyde

Solution 1: (b) Acetophenone

Question 2: Oxidation states of scandium are_____. [1]

- (a) +1, +2 (b) +1, +3 (c) +2, +3 (d) +3, +4

Solution 2: (c) +2, +3

Question 3: In van Arkel method for refining zirconium or titanium, the halogen used is_____. [1]

- (a) Fluorine (b) Chlorine (c) Bromine (d) Iodine

Solution: (d) Iodine.

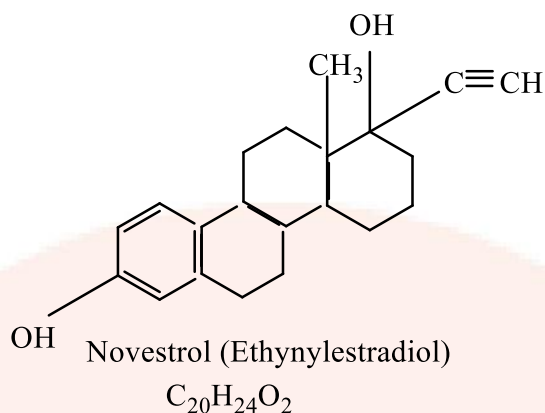
Question 4: A system absorbs 6kJ of heat and does 1.5kJ of work on its surroundings. The change in internal energy is _____. [1]

- (a) -7.5 kJ (b) -4.5 kJ (c) +4.5 kJ (d) +7.5kJ

Solution 4: (c) +4.5 kJ

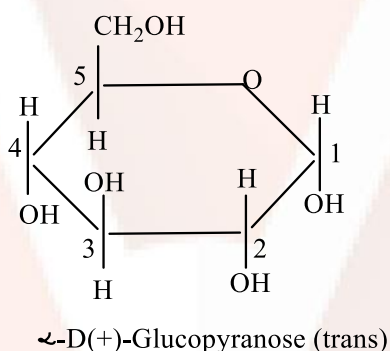
Question 5: Write the molecular formula of novestrol.

Solution 5:



Question 6: Write the number of hydroxyl groups present in α -D(+)-Glucopyranose(trans.) [1]

Solution 6:



Question 7: What is Nessler's reagent? [1]

Solution 7: Nessler's reagent is an alkaline solution of K_2HgI_4 .

Question 8: What is the ratio of octahedral holes to the number of anions in hexagonal closed packed structure? [1]

Solution 8: 1:1

Section – B

Question 9: What are ethers? How are they classified? [2]

Solution 9: Ethers are alkoxyderivatives of alkane

Ethers are classified into two groups as,

- (i) Simple or symmetric ethers: The ethers, in which both the alkyl or aryl group attached to oxygen atom are same, are called simple ethers.
Example, $\text{CH}_3\text{-O-CH}_3$ dimethyl ether.
- (ii) Mixed or unsymmetrical ethers: The ethers, in which the two alkyl or aryl groups attached to oxygen atom are different, are called mixed ether.
Example, $\text{CH}_3\text{-O-C}_2\text{H}_5$ dimethyl ether.

Question 10. What are antacids? Write the main constituents of Dettol. [2]

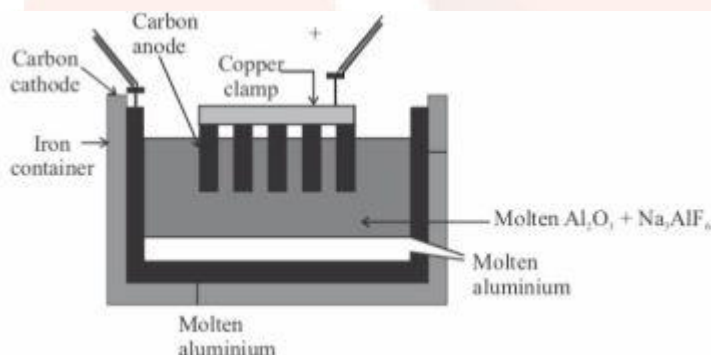
Solution 10: Antacids are the chemical which neutralize excess acidity in the stomach.

Constituents of Dettol – chloroxylenol and terpeniol.

Question 11 Draw a neat, labeled diagram of electrolytic cell for the extraction of aluminium.

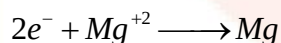
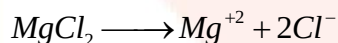
Solution 11:

Electrolytic cell for the extraction of aluminium



Question 12: How many faradays of electricity are required to produce 6g of Mg from MgCl_2 ?

Solution 12:



$$\text{Mole ratio} = \frac{1}{2}$$

$\therefore 1F$ / electricity deposits = 12gm Mg

$\therefore 6\text{gm Mg}$ requires = 0.5F

OR

The molar conductivity of 0.05 M BaCl_2 solution at 25°C is $223\Omega\text{cm}^2\text{mol}^{-1}$. What is its conductivity?

Solution:

$$C = 0.05M$$

$$\lambda_m = 223\Omega^{-1}cm^2mol^{-1}$$

$$\lambda_m = \frac{1000k}{C}$$

$$\therefore k = \frac{\lambda_m \times C}{1000}$$

$$= \frac{223 \times 0.05}{1000}$$

$$= 0.01115Scm^{-1}$$

Question 13: Derive van't Hoff general equation.

Solution 13:

Van't Hoff deduced by combining of Boyle's and Charle's law.

$$\pi \propto \frac{T}{V}$$

$$\Rightarrow \pi V = KT$$

π = osmotic pressure

V = Volume of solution

R = gas constant

T = absolute temprature

Question 14: Write the condition for maximum work done by the system.

Solution 14:

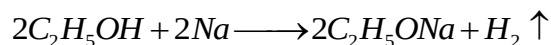
Conditions for maximum work:

- (i) Any change taking place in it is thermodynamically reversible.
- (ii) The change in the system takes place in an infinite number of steps.
- (iii) The driving force of the change is infinitesimally greater than the opposite force.
- (iv) The system is in mechanical equilibrium with its surrounding.

Question 15: Write balanced chemical equation for the following.

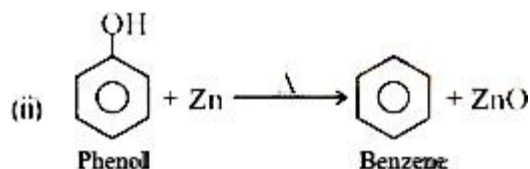
- (i) **Action of sodium metal on ethanol.**

Sol:



- (ii) **Action of zinc dust on phenol.**

Sol:

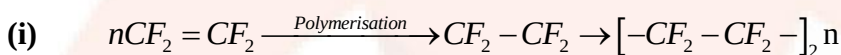


SECTION C

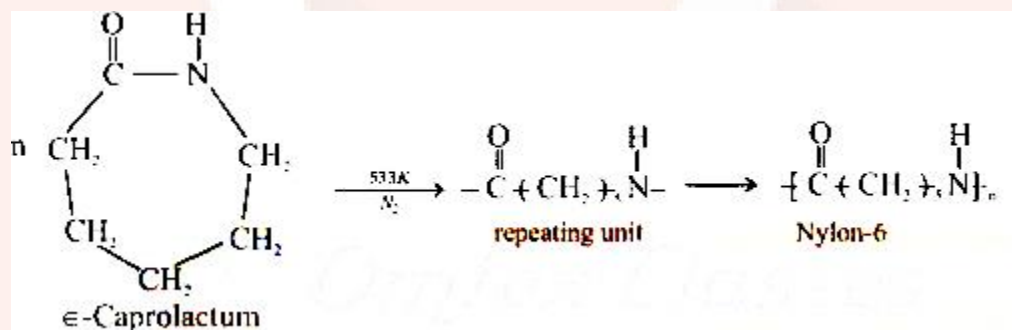
Question 16: Write the chemical reaction to prepare the following polymers.

- (i) Teflon
- (ii) Nylon-6
- (iii) Dextran

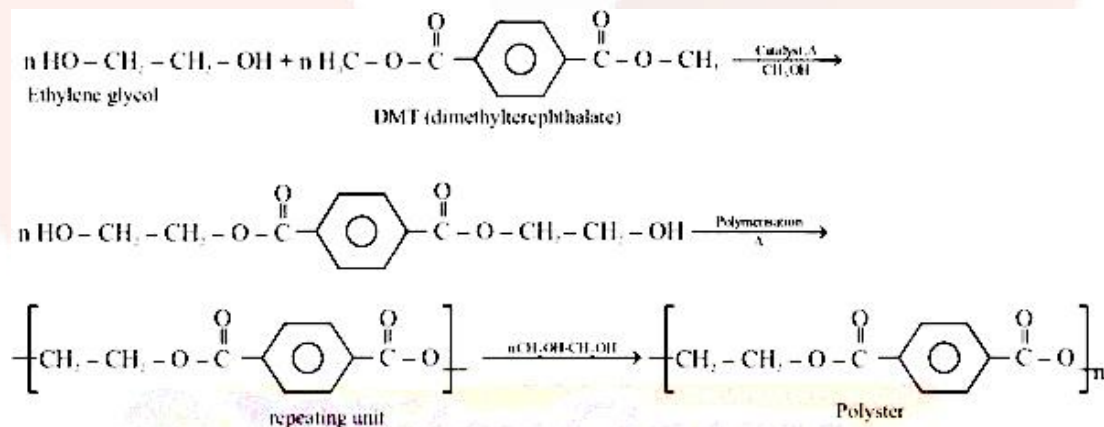
Solution 16:



(ii)



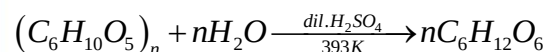
(iii)



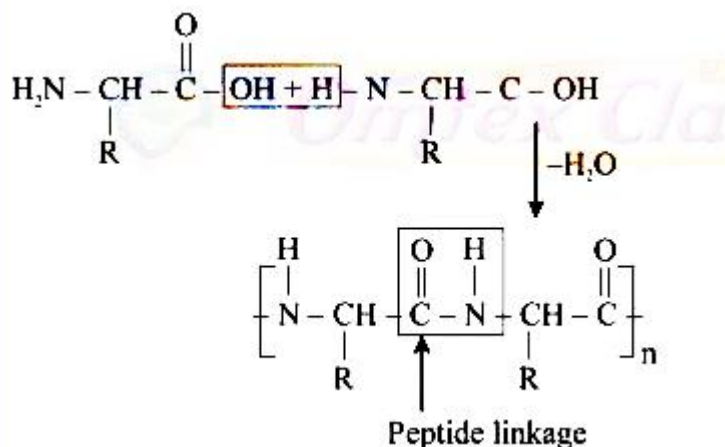
Question 17: how is glucose prepared by commercial method? How is peptide linkage formed?

Solution 17:

Commercially glucose is prepared using starch. Starch is boiled with dil H_2SO_4 at 393K. it goes on hydrolysis process and gives glucose



Peptide linkage is formed when the amino acids undergo condensation with loss of water.

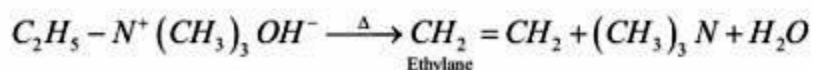
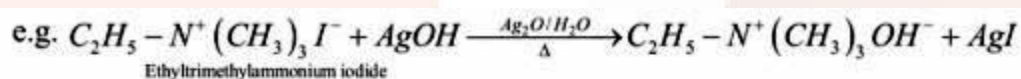


Question 18: Write a short note on ethyl bromide.

Solution 18:

If the tetra alkyl ammonium halide is heated with moist silver hydroxide, it gives quaternary ammonium hydroxide.

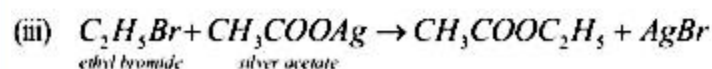
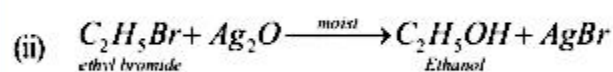
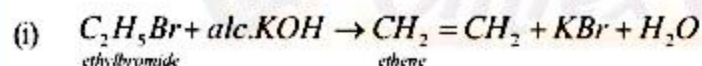
Quaternary ammonium hydroxide on strong heating undergoes β elimination to give alkenes. Therefore, this reaction is known as Hoffman elimination.



Question 19: What is the action of the following on ethyl bromide?

- (I) Alcoholic solution of potassium hydroxide
- (II) Moist silver oxide
- (III) Silver acetate

Solution 19:



Question 20: What is effective atomic number? Calculate effective atomic number of copper ($Zn = 29$) in $(Cu(NH_3)_2)^{2+}$

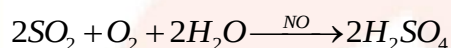
Solution 20:

EAN is the total number of electrons around central metal ion present in a complex.

$$[Cu(NH_3)_2]^{2+} \rightarrow 29 - 2 + 8 \rightarrow 35$$

Question 21: Write the chemical reaction for different steps in the manufacture of sulphuric acid by lead chamber process. Draw the structure of phosphorus pentachloride.

Solution 21:



And this reaction can be written as the sum of the following reaction taking place in different steps.

- (I) $2H_2O + NO \rightarrow HNO_3 + 3H^+$
- (II) $S + O_2 \rightarrow SO_2$ (oxidation of sulphur)
- (III) $2HNO_3 + 2SO_2 \rightarrow H_2O + NO + NO_2 + 2SO_3$
- (IV) $SO_3 + H_2O \rightarrow H_2SO_4$

Question 22: Write Arrhenius equation. Derive an expression for temperature variations.

Solution 22:

Arrhenius equation, $K = A \cdot e^{\frac{-E_a}{RT}}$ for temperature variation.

$$\log_{10} k = \log_{10} A - \frac{E}{2.303RT}$$

At two different temperature T_1 and T_2

$$\log_{10} k_1 = \log_{10} A - \frac{E}{2.303RT_1} \quad \dots(1)$$

And

$$\log_{10} k_2 = \log_{10} A - \frac{E}{2.303RT_2} \quad \dots\dots(2)$$

Subtract eq (1) from eq(2)

$$\log_{10} k_1 = \log_{10} A - \frac{E}{2.303RT_1}$$

Question 23: Define electrochemical series. Write its application.

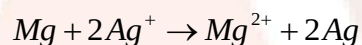
Solution 23:

EMF series is defined as the arrangement of electrode with the electrode half reaction in order of decreasing standard potential.

Its applications are as follows:

- (1) Relative strength of oxidizing agents in terms of E^0 values.
The left side of half reaction is oxidizing agents. The E^0 value is measure of the tendency of the species to be reduced.
- (2) The standard electrode potential can be used to determine the spontaneity of redox reaction. E_{cell}^0 for the overall cell reaction can be calculated from the E^0 values, listed in the series, for the half reaction which constitute over all cell reaction.

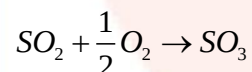
For example we have to decide whether Mg is oxidized by Ag^+



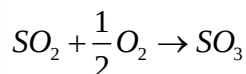
Now, the half reactions for $Ag^+ | Ag$ and $Mg^{2+} | Mg$

- (i) $Ag^+ + e^- \rightarrow Ag$ $E^0 = 0.8V$
- (ii) $Mg^{2+} + 2e^- \rightarrow Mg$ $E^0 = -2.37$
 $Mg \rightarrow Mg^{2+} + 2e^-$ $E^0 = -(-2.37V)$
- (iii) $2Ag^+ + 2e^- \longrightarrow 2Ag$ $E^0 = 0.8V$
 $Mg \longrightarrow Mg^{2+} + 2e^-$ $E^0 = 2.37V$
 $2Ag^+ + Mg \longrightarrow 2Ag + Mg^{2+}$ $E_{cell}^0 = 3.17V$

Question 24: Calculate the work done in the following reaction at $50^\circ C$. State whether work is done on the system or by the system.



Solution 24:



$$\Delta n = (\text{moles of gaseous product}) - (\text{mass of gaseous reactant})$$

$$= 1 - \left(1 + \frac{1}{2}\right)$$

$$= \frac{-1}{2}$$

$$W = -\Delta nRT$$

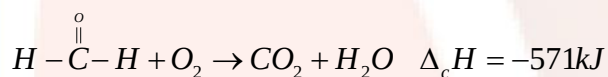
$$= -\left(\frac{-1}{2} \times 8.314 \times 323\right)$$

$$= 1342.7 J$$

OR

The standard enthalpy of combustion of formaldehyde $\Delta_c H^0 = -571 kJ$. How much heat will be evolved in the formation of 22g of CO_2 ?

Solution:



for 44g of CO_2 ,

$$\Delta H = -671 kJ$$

$\therefore$ 22g of CO_2 ,

$$\Delta H = \frac{22 \times -571}{44}$$

$$= -285.5 kJ$$

Question 25: Define the following terms:

- (i) Isotonic solution
- (ii) Hypertonic solution
- (iii) Hypotonic solution

Solution 25:

- (i) **Isotonic solution:-** Two or more solutions having same osmotic pressure.
- (ii) **Hypertonic solution:-** The solution having higher osmotic pressure than other.
- (iii) **Hypotonic solution:-** The solution having lower osmotic pressure than other.

Question 26: Define Anisotropy. Distinguish between crystalline solids and amorphous solids.

Solution 26:

The ability of crystalline to change value of physical properties when measured in different direction.

The difference between crystalline solids and Amorphous solids are as follows:

Crystalline solids	Amorphous solids
1. The solids have definite characteristic shape due to orderly regular long range arrangement.	a. The solids gave irregular shape due to oderly arrangement of constituent particles.
2. These are true solids.	b. These are pseudo solids or super cooled liquid.
3. It may or may not be isotropic.	c. It is isotropic like liquid.
4. Heat of fusion is definite and depends upon arrangement of particles in the crystalline solids.	d. Heat of fusion is not definite.

SECTION D

Question 27: (A) Write the stracture of Phenylmethamine.

(B) Write chemical equation for the following:

(i) Gatterman- Koch formylation

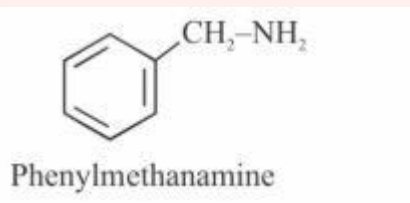
(ii) Rosenmund reduction

(iii)Fisher esterification reduction

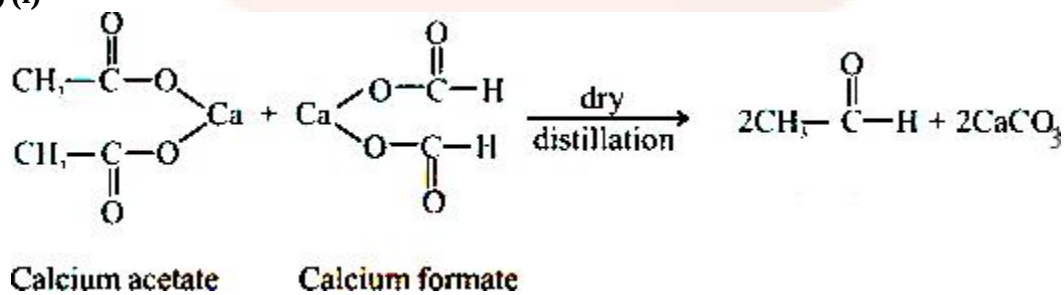
(iv)Hell- Vohlard- Zelinsky reaction

Solution 27:

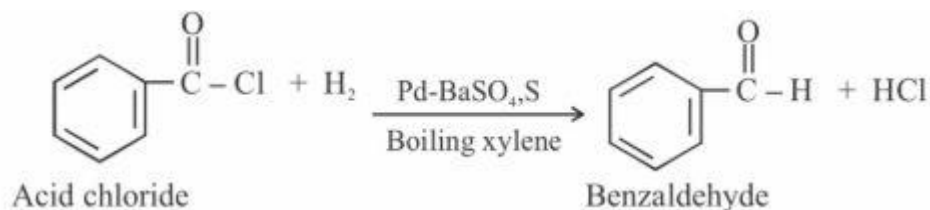
(A)



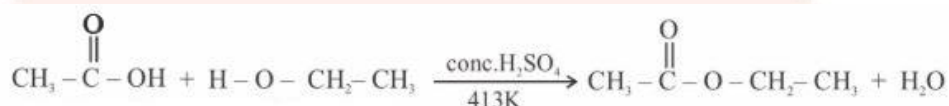
(B) (i)



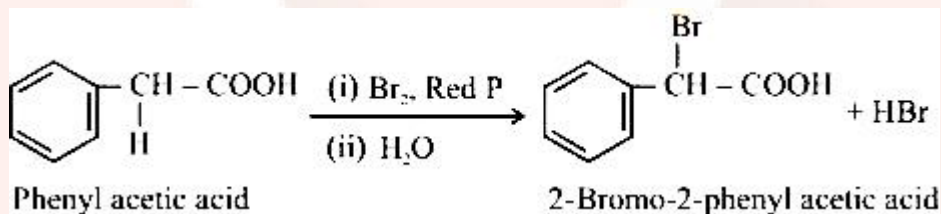
(ii)



(iii)



(iv)/



OR

(A) What are amines?

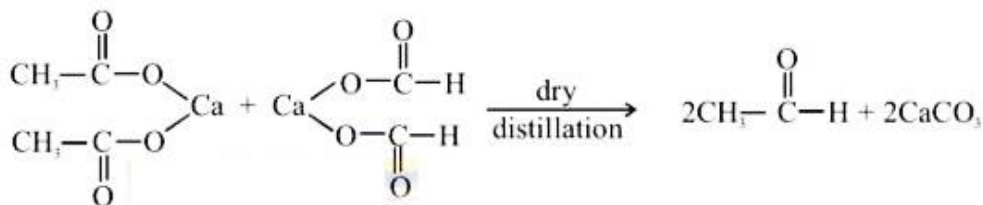
(B) How will you convert

- (i) Calcium acetate to acetaldehyde
- (ii) Acetone to acetone cyanohydrine
- (iii) Sodium acetate to methane
- (iv) Benzoic acid to m- bromobenzoic acid?

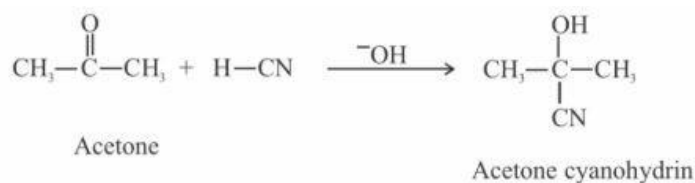
Solution:

(A) Amines are the organic derivatives of ammonia in which one, two or all the three hydrogen atoms attached to nitrogen are replaced by equivalent number of same or different group.

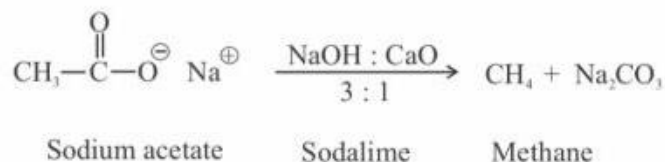
(B) (i)



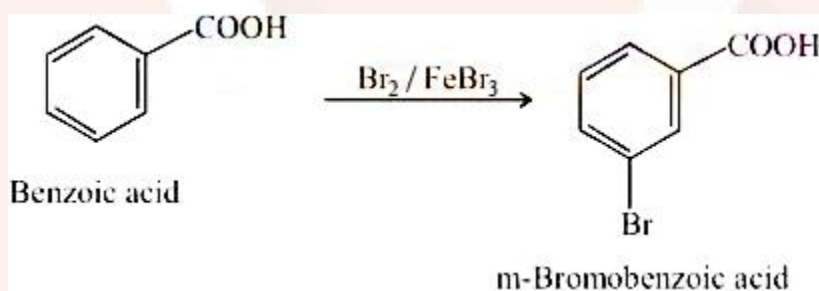
(ii)



(iii)



(iv)



Question 28:

(A) Define Enantiomers.

(B) How is potassium dichromate prepared from chrome iron ore?

Solution 28:

(A) Stereoisomers which are non- superimposable mirror images of each other and rotate the plan of polarized light through the same angle but in opposite direction.

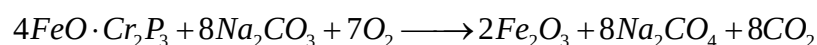
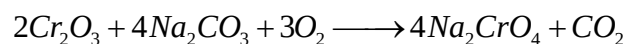
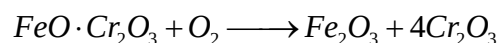
(B)

Step 1:

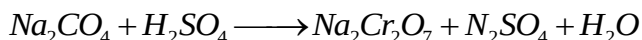
Power chromite ore is concentrated by washing with current of water in a hydraulic classifier.

Lighter gangue is carried with water.

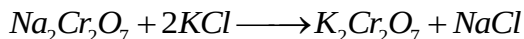
Steps 2:



Steps 3:



Steps 4:



OR

(A) What is Grignard reagent?

(B) Explain the position of actinoids in the periodic table.

What is the action of sulphur in lanthanoids

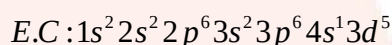
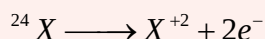
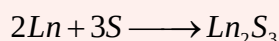
Calculate the magnetic moment of divalent ion in aqueous solution its atomic number is 24

Solution:

(A) Grignard reagent is an organometallic compound in which the divalent magnesium is directly linked to an alkyl group and a halogen atom.

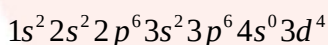
(B) Actinoids have a general electronic configuration $5f^{1-14}6d^{0-1}7s^2$.

Placed at the bottom of periodic table.



in

+2 state :



$\therefore$ Number of unpaired electron = 4

$$u = \sqrt{n(n+2)}$$

$$= \sqrt{4(4+2)}$$

$$= \sqrt{24}$$

$$= 4.89 B.M$$

Question 29:

(A) The rate of the first order reaction $A \rightarrow B$ is $5.4 \times 10^{-6} \text{ Ms}^{-1}$ when $[A]$ is 0.3M.

Calculate the rate reaction.

(B) Explain the following properties of group 16 elements:

(i) Electro negativity

- (ii) Melting and boiling points
- (iii) Metallic and boiling points
- (iv) Allotropy

Solution 29:

(A)

$$\text{Rate} = k[A]$$

$$5.4 \times 10^{-6} = k \times 0.3$$

$$k = \frac{5.4 \times 10^{-6}}{0.3}$$

$$= 18 \times 10^{-6} \text{ s}^{-1}$$

(B)

- (i) The elements of group 16 have higher value of electronegativity than the corresponding elements of group 15. Oxygen is the second most electronegative element. The electronegativity decreases on going down the group.
- (ii) The boiling point and the melting point increase with the increase of atomic number.
- (iii) The first four elements oxygen, sulphur, selenium and tellurium are non metals. The non metallic character is stronger in O and S are weaker in Se and Te.
- (iv) All the elements of the group exhibit allotropy. E.g. O_2 and O_3 . Sulphur exists in a number of allotropic forms of which yellow ortho- rhombic, α and β - monoclinic forms are most important.

OR

(A) The half-life period of a first order reaction is 6.0h. calculate the rate constant.

(B) What are oxides?

Write different types of oxides with one example each.

Solution:

(A)

$$t_{\frac{1}{2}} = 6.0 \text{ hr}$$

$$= \frac{0.693}{k}$$

$$= \frac{0.693}{t_{\frac{1}{2}}}$$

$$= \frac{0.693}{6 \times 60 \times 60}$$

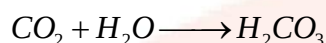
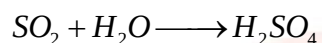
$$= 0.000032 \text{ s}^{-1}$$

$$= 3.2 \times 10^{-5} \text{ s}^{-1}$$

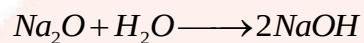
(B)

The binary compounds of oxygen with other elements are called oxides. We have studied that oxygen combines with metals and non-metals to form their respective binary oxides.

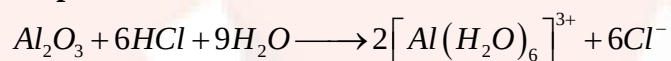
1. **Acidic oxide:** The oxides which combine with water to give acids are called acidic oxides.



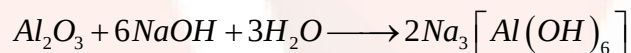
2. **Basic oxides:** The oxide which combine with water to give basic solution are called basic oxides.



3. **Amphoteric oxides:** The oxides which show acidic as well as basic.



(Basic)



(Acidic)